FRDM-MW30-S

Table of Content

Page 1	Title Page
Page 2	Block Diagram
Page 3	System PWR
Page 4	PWR Management
Page 5	MW30
Page 6	HMI/Sensor/Memory
Page 7	CAN/LIN
Page 8	Arduino/MikroE/Debug
Page 9	MCU_LINK_USB
Page 10	MCU_LINK_DEBUG
Page 11	MCU_LINK_CURRENT
Page 12	MCU_LINK_VCOM_SIO
Page 13	
Page 14	
Page 15	
Page 16	
Page 17	

NOTES

1. Unless Otherwise Specified:
   All resistors are in ohms, 5%, 1/8 Watt
   All voltages are DC

2. Interrupted lines coded with the same letter or letter combinations are electrically connected.

3. Device type number is for reference only. The number varies with the manufacturer.

4. Special signal usage:
   _B Denotes - Active-Low Signal
   <> or [] Denotes - Vectored Signals

5. Interpret diagram in accordance with American National Standards Institute specifications, current revision, with the exception of logic block symbology.

Revision History

Rev. Code	Date	By	Description
A	2025-12-19	Shawn Shi	Initial Release

Jumper Setting

REF DES	JUMPER(DEFAULT)	PAGE NAME
J2,J4,J32,JP1	1-2	03. System PWR
J5	OPEN	03. System PWR
J8,J9,J10,J11,JP2,JP3	1-2	04. PWR Management
J13,J14,J15	1-2	06. HMI/Sensor/Memory
JP21,JP22	1-2	07. CAN/LIN
J17	OPEN	07. CAN/LIN
JP5	1-2	08. Arduino/MikroE/JTAG
JP8,JP20	1-2	09. MCU_LINK_USB
JP6,JP7,JP9	OPEN	09. MCU_LINK_USB
JP12	1-2	10. MCU_LINK_DEBUG
JP10,JP11	OPEN	10. MCU_LINK_DEBUG
JP13,JP14,JP15,JP16	1-2	11. MCU_LINK_CURRENT
JP17	1-2	12. MCU_LINK_VCOM_SIO
JP18,JP19	2-3	12. MCU_LINK_VCOM_SIO

DNP Setting

REF DES	ASSY_OPT	PAGE NAME
C15,J3,J6,J7,R1,R2,R5,U4	DNP	03. System PWR
C19,C20,C23	DNP	04. PWR Management
C43,C44,C45,C46,C47,C51,C136, C137,C138,C139	DNP	05. MW30
C53,C54,R62,R73,R76,R77,R90, R91,R92,R93	DNP	06. HMI/Sensor/Memory
J33	DNP	07. CAN/LIN
D10,R140,R141,R142,R144,R147, R149,TP9	DNP	08. Arduino/MikroE/JTAG
C94,C95,J29,J30,R169,R181, R182	DNP	09. MCU_LINK_USB
D21,D22,D23,J31,R188	DNP	10. MCU_LINK_DEBUG
R208,R221	DNP	11. MCU_LINK_CURRENT

NXP

ICAP Classification: CP: ___ IUO: X PUBL: __

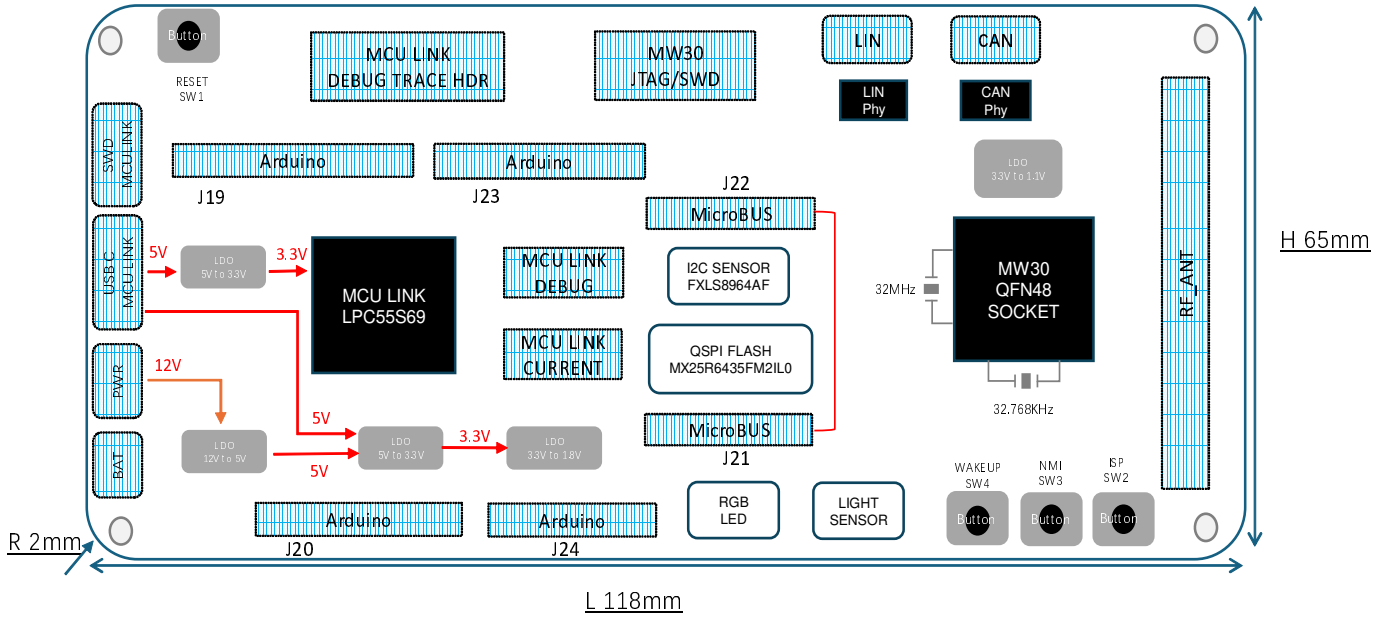
Drawing Title: FRDM-MW30-S

Page Title: Title Page

Size	Document Number	Rev
C	SCH-95995 PDF: SPF-95995	A

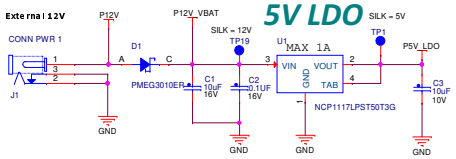
Date: Friday, December 19, 2025 Sheet 1 of 12

FRDM-MW30-S

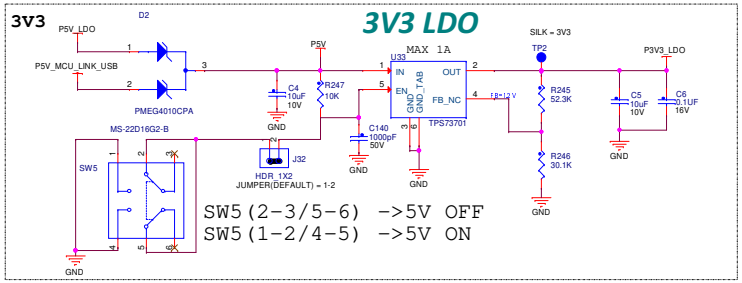


ICAP Classification:	CP: ___ IUO: X PUBI: __	
Drawing Title:	FRDM-MW30-S	
Page Title:	Block Diagram	
Size C	Document Number SCH-95995 PDF: SPF-95995	Rev A
Date: Friday, December 19, 2025	Sheet 2 of 12	

CAN/LIN Power

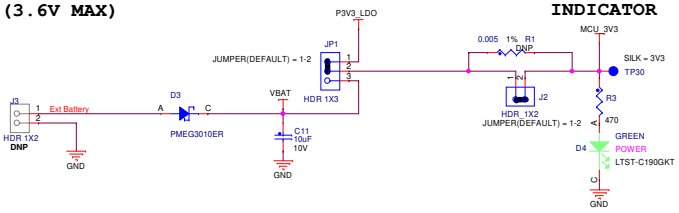


3V3

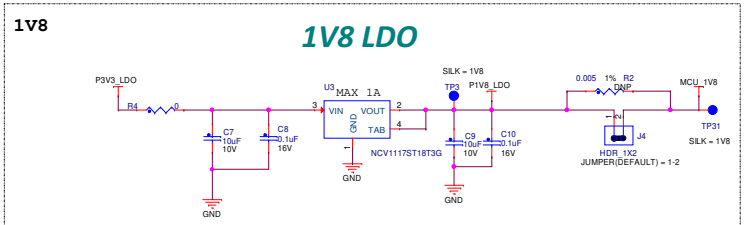


EXT BATTC (3.6V MAX)

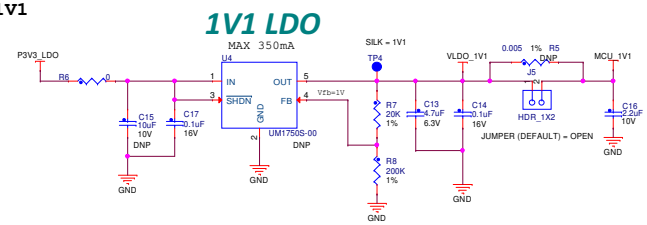
POWER ON INDICATOR



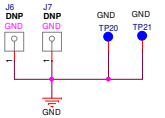
1V8



1V1



GND Test Points



Note:
Two ground test points are placed on the left and right sides of the board to facilitate testing

Mounting Holes



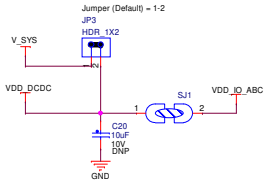
Note:
The mechanical mounting holes are placed on the four vertices of the board

ICAP Classification:	CP: ___ IUO: X PUBI: __	
Drawing Title:	FRDM-MW30-S	
Page Title:	System Power	
Size: C	Document Number: SCH-95995 PDF: SPF-95995	Rev: A
Date: Thursday, December 18, 2025	Sheet 3 of 12	

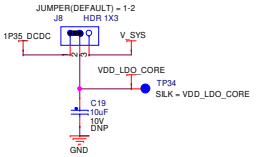
POWER OPTIONS	VDD_IO/VDD_DCDC [1.8V-3.6V]		VDD_LDO_CORE [1.25V-3.6V]	VDD_ANA [1.71V-3.6V]	VDD_VREFH [1.71V-3.6V]	VDD_RF [1.175V-3.6V]
	JP2	JP3	J8	J9	J10	J11
DCDC Mode	1-2	1-2	1-2	1-2	1-2	1-2
LDO Mode	1-2	1-2	2-3	1-2	1-2	2-3

Power Options

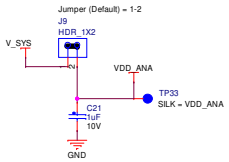
VDD_IO/VDD_DCDC



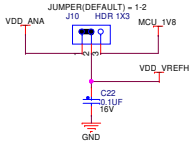
VDD_LDO_CORE



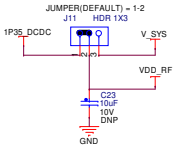
VDD_ANA



VDD_VREFH

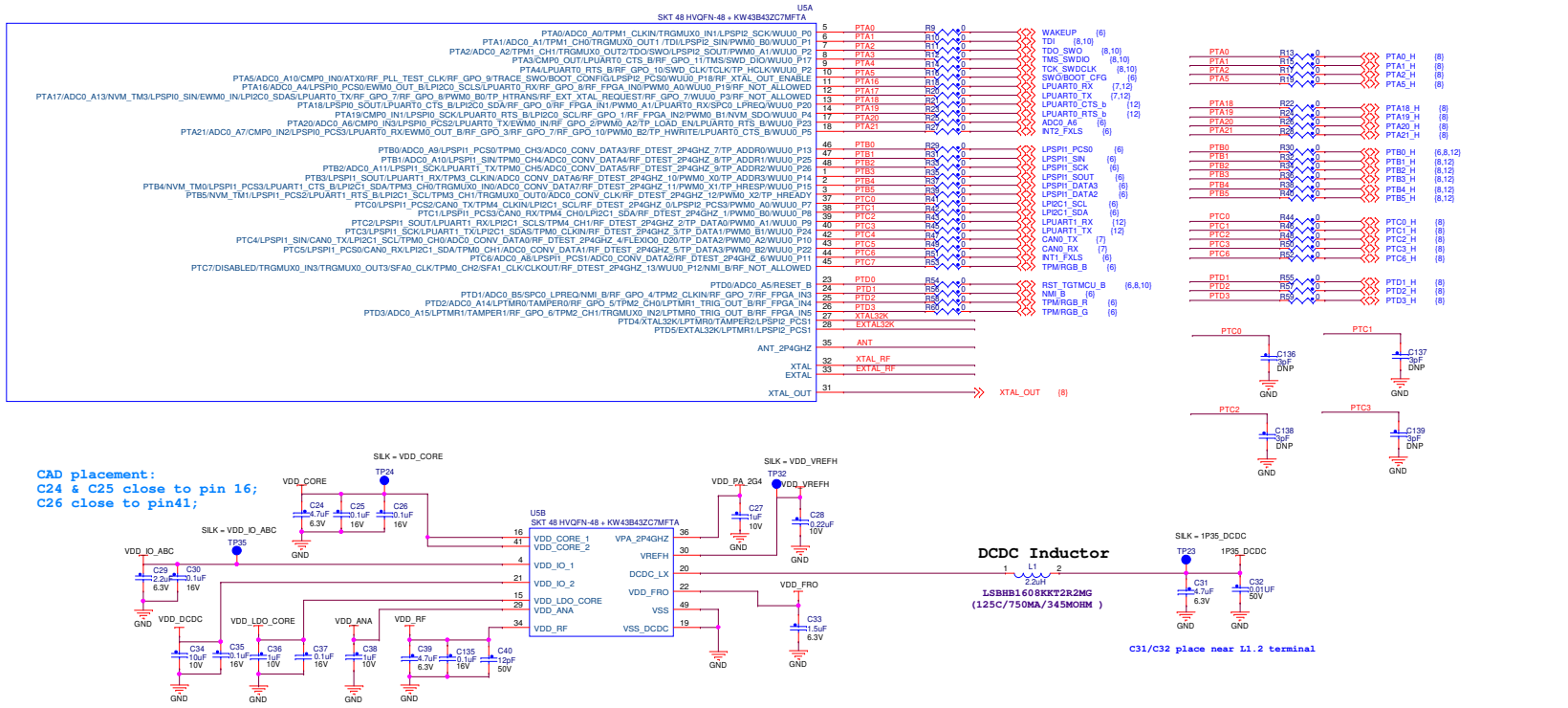


VDD_RF



ICAP Classification:	CP: ___ IUO: X PUBI: __	
Drawing Title:	FRDM-MW30-S	
Page Title:	PWR Management	
Size: C	Document Number: SCH-95995 PDF: SPF-95995	Rev: A
Date: Thursday, December 18, 2025	Sheet 4 of 12	

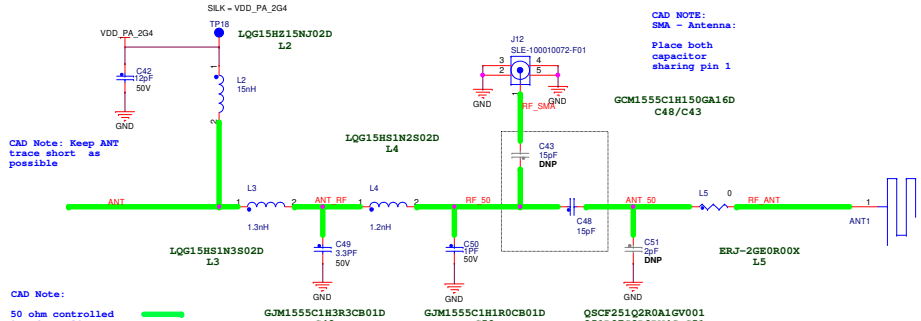
MW30



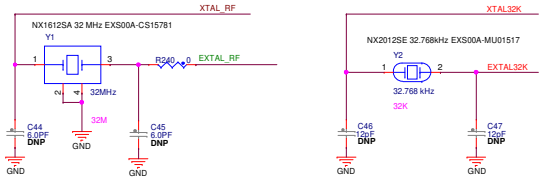
CAD placement:
C24 & C25 close to pin 16;
C26 close to pin41;

C31/C32 place near L1.2 terminal

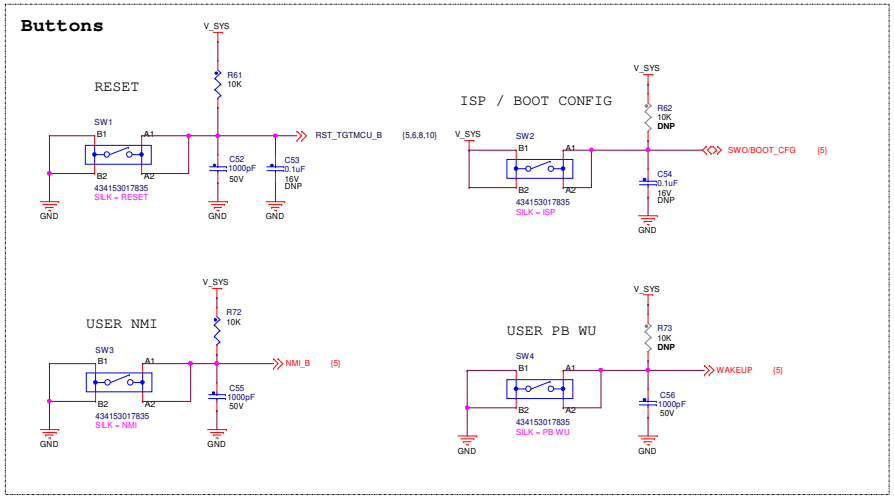
RF & ANTENNA

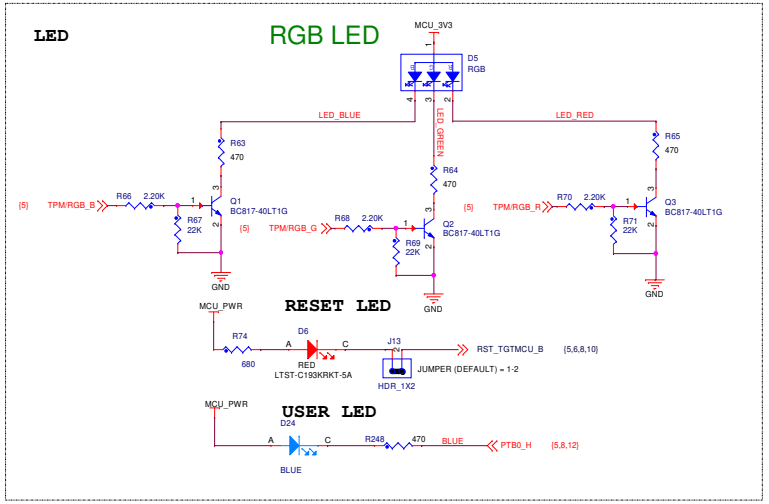


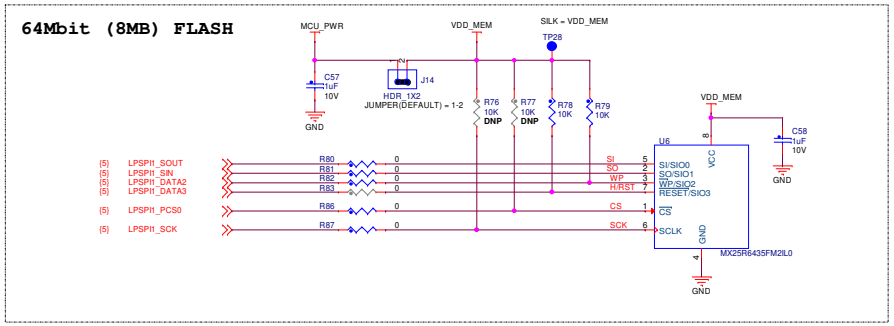
CLOCK SECTION

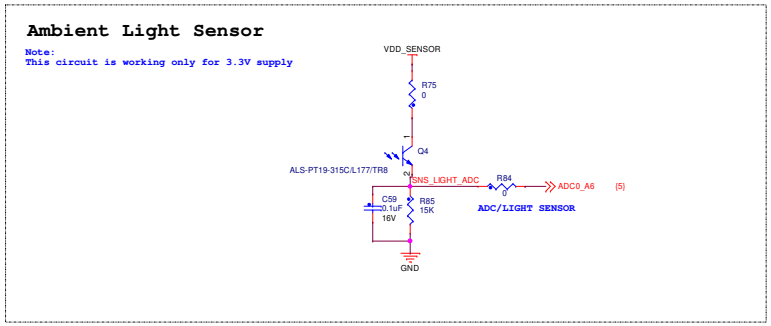


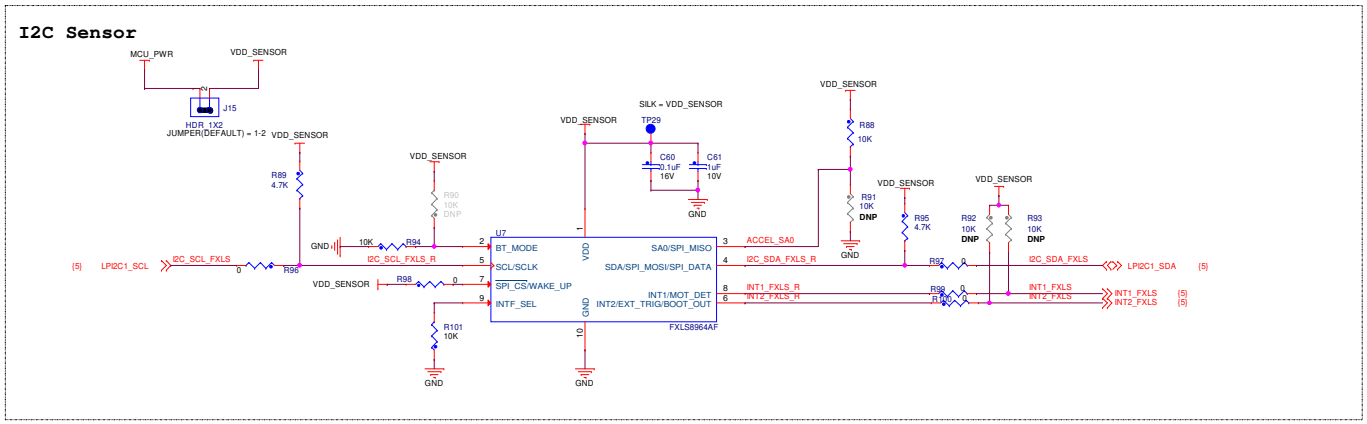
ICAP Classification:	CP: ___	IUO: X	PUBI: ___
Drawing Title:	FRDM-MW30-S		
Page Title:	MW30		
Size C	Document Number SCH-95995 PDF: SPF-95995		Rev A
Date: Thursday, December 18, 2025	Sheet 5 of 12		











NXP

ICAP Classification: CP: IUO: X PUBI:

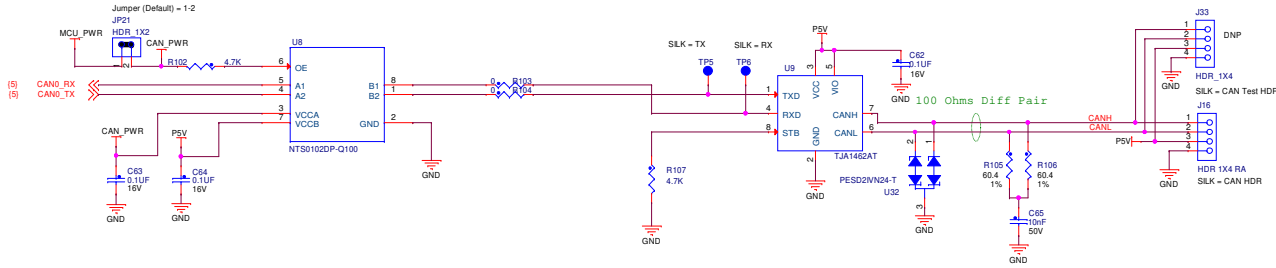
Drawing Title: **FRDM-MW30-S**

Page Title: **HMI/Sensor/Memory**

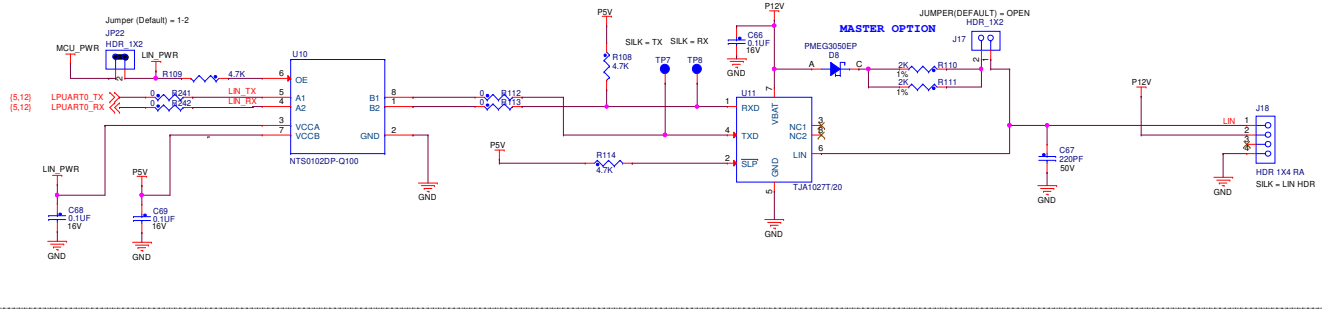
Size	Document Number	Rev
C	SCH-95995 PDF: SPF-95995	A

Date: Thursday, December 18, 2025 Sheet 6 of 12

CAN

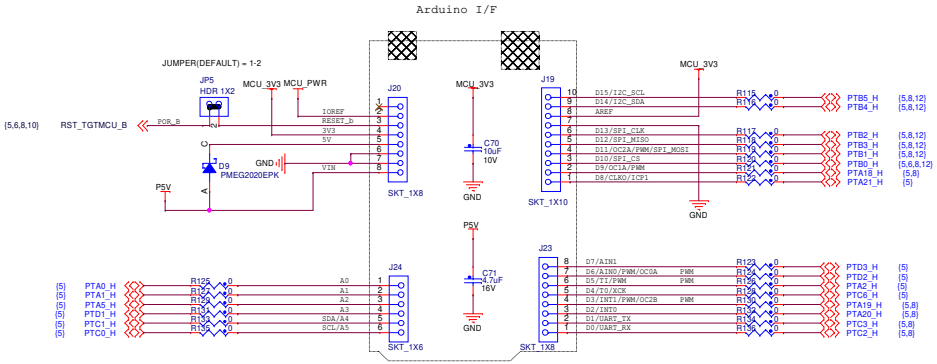


LIN



ICAP Classification:	CP:	IUO: X	PUBI:
Drawing Title:	FRDM-MW30-S		
Page Title:	CAN/LIN		
Size C	Document Number SCH-95995 PDF: SPF-95995		Rev A
Date: Thursday, December 18, 2025	Sheet 7 of 12		

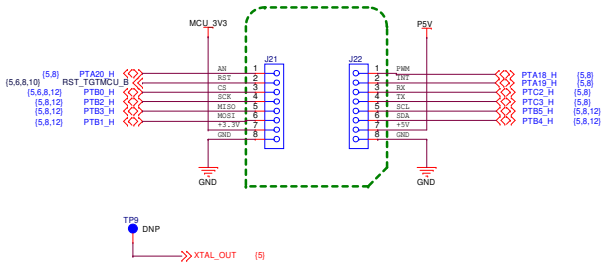
Arduino



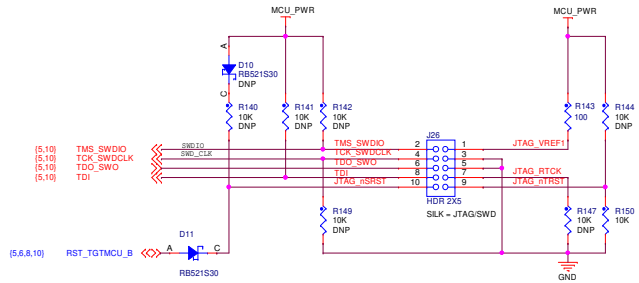
The connectors used are 2.54mm sockets

NOTE: Arduino shield used should not exceed 200 mA power requirements through 3V3.

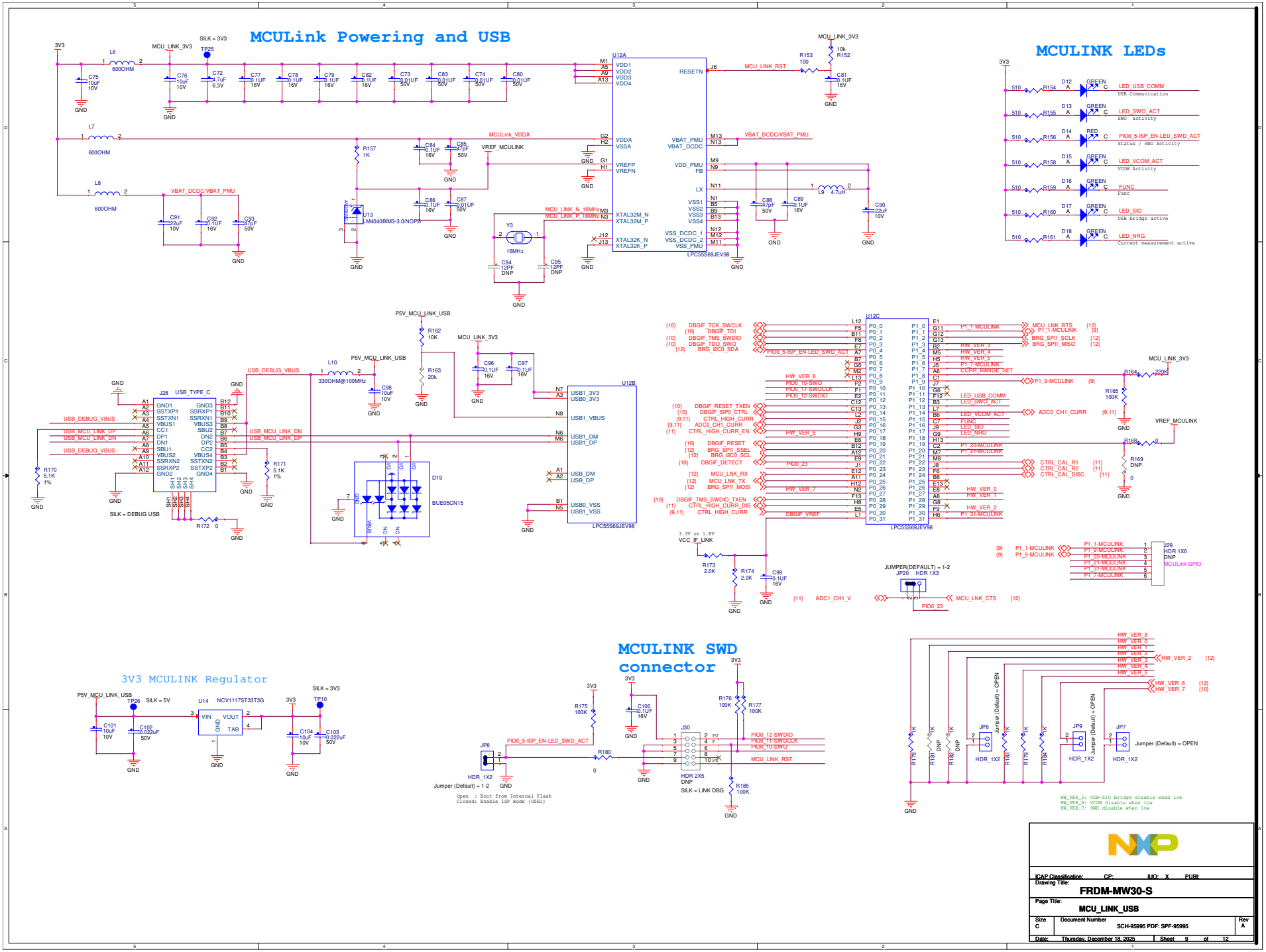
Mikroe



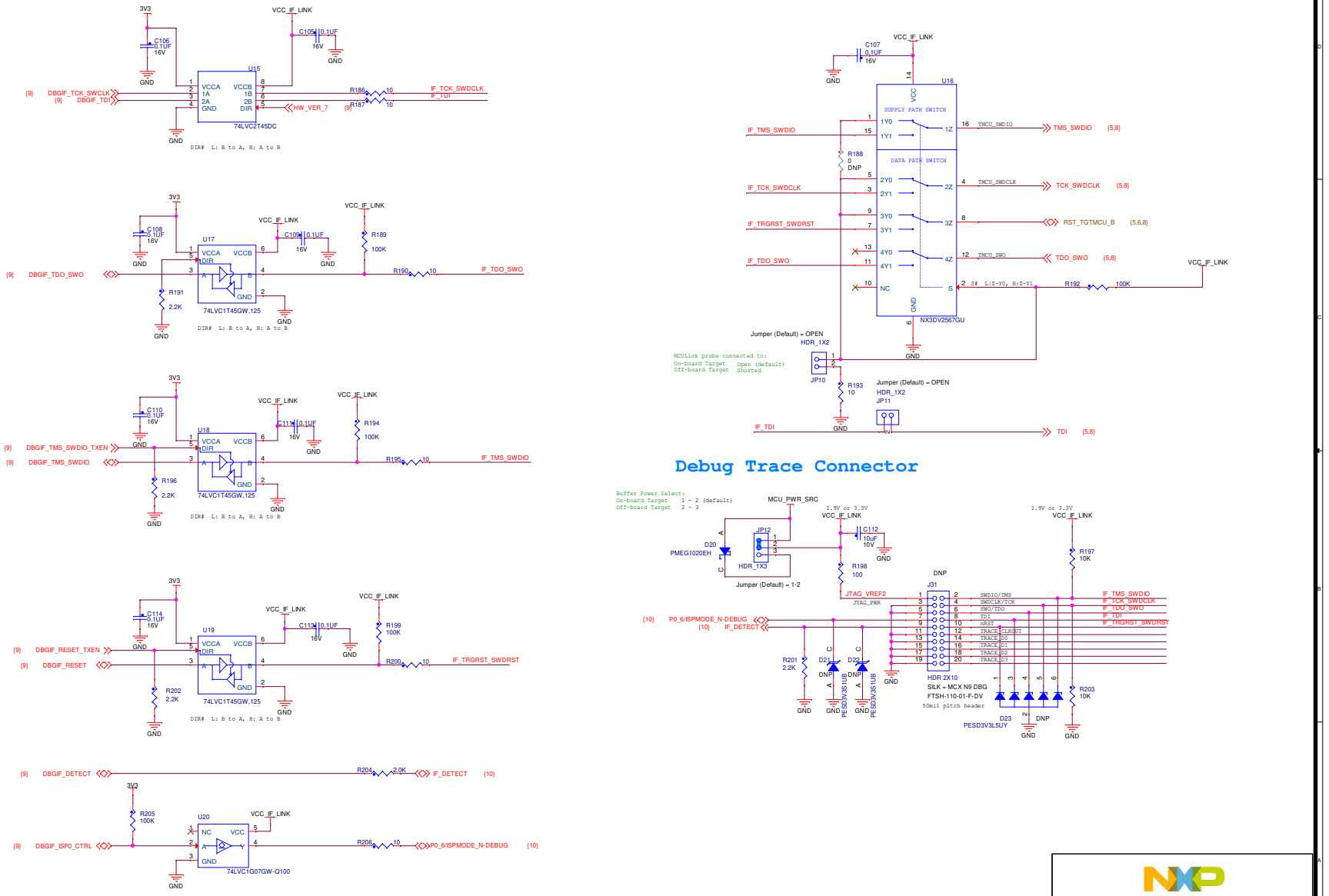
JTAG/SWD (DEFAULT)



ICAP Classification:	CP:	IUO: X	PUBI:
Drawing Title:	FRDM-MW30-S		
Page Title:	Arduino/MikroE/Debug		
Size C	Document Number SCH-95995 PDF: SPF-95995		Rev A
Date: Friday, December 19, 2025	Sheet 8 of 12		


MCULink Powering and USB
MCULINK LEDs
3V3 MCULINK Regulator
MCULINK SWD connector
FRDM-MW30-S
MCU_LINK_USB
SCH-95995 PDF: SPF-95995
Thursday, December 18, 2025

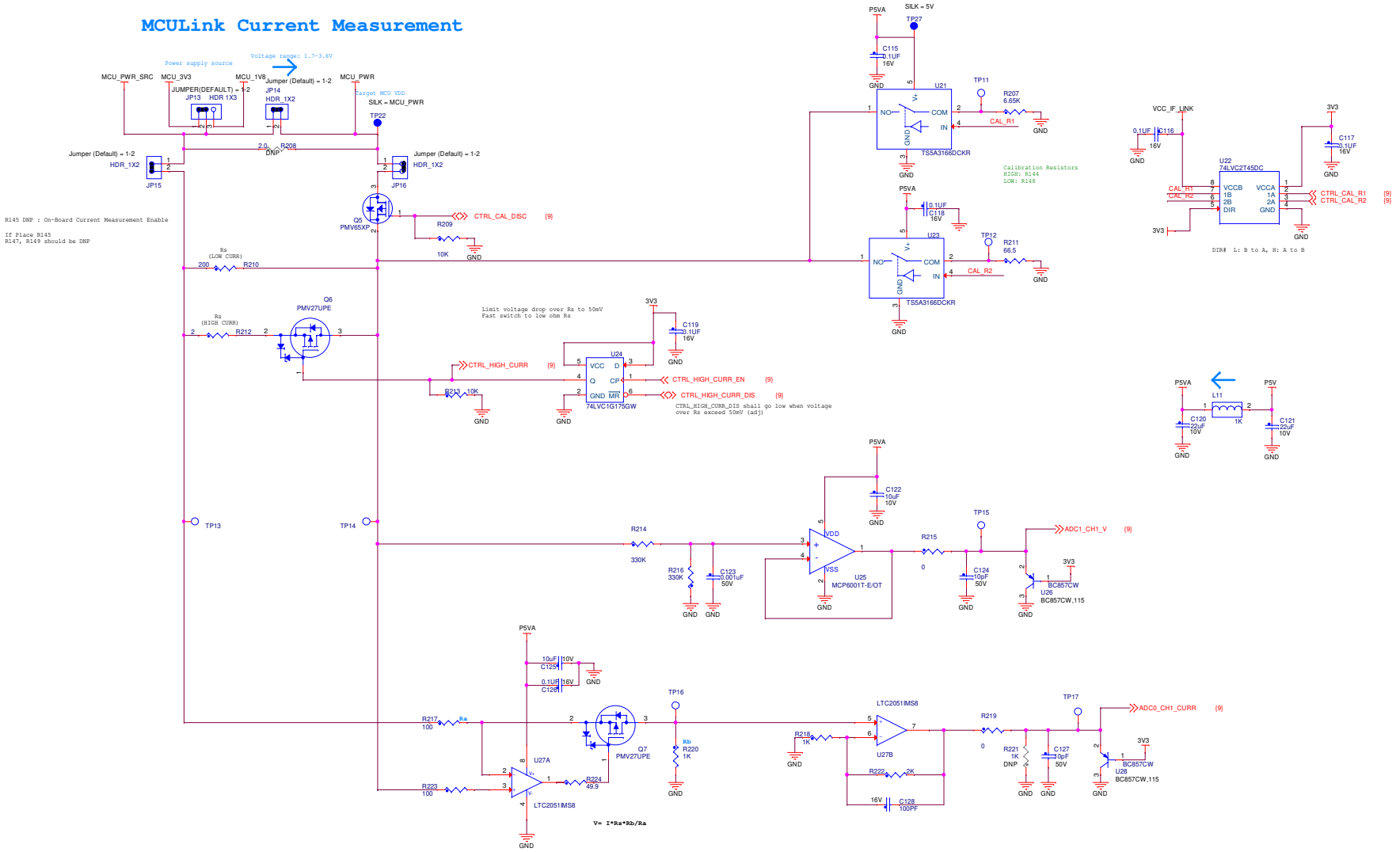
MCULink Debug Interface



Debug Trace Connector

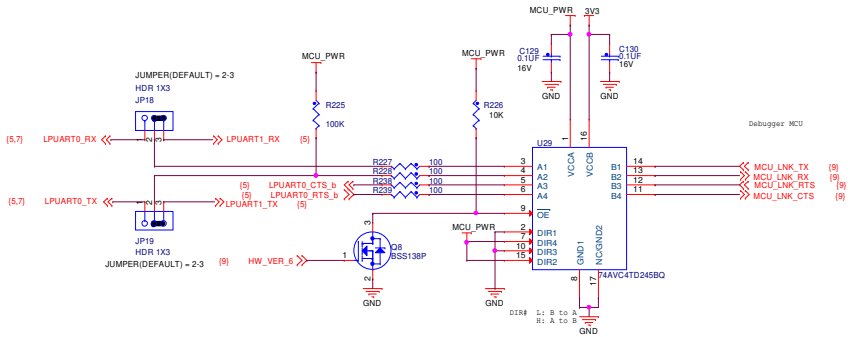
ICAP Classification:	CP:	IUO: X	PUBI:
Drawing Title:	FRDM-MW30-S		
Page Title:	MCU_LINK_DEBUG		
Size C	Document Number SCH-95995 PDF: SPF-95995		Rev A
Date: Thursday, December 18, 2025	Sheet 10 of 12		

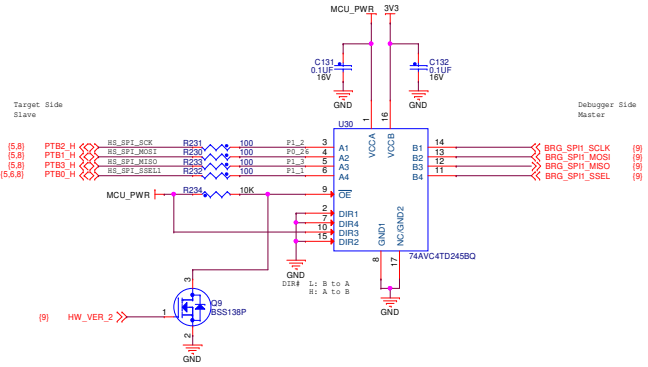
MCULink Current Measurement

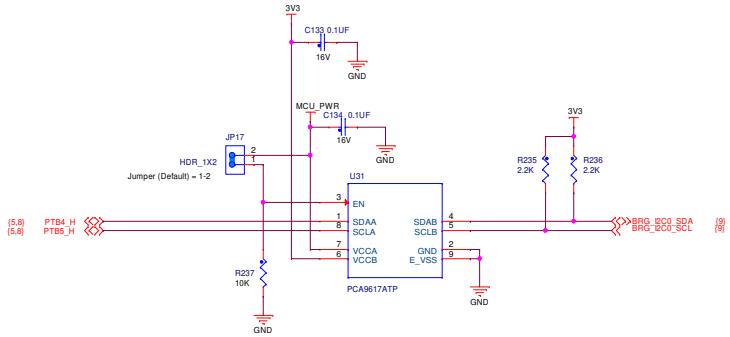


ICAP Classification:	CP:	IUO: X	PUBI:	
Drawing Title:	FRDM-MW30-S			
Page Title:	MCU_LINK_CURRENT			
Size C	Document Number SCH-95995 PDF: SPF-95995			Rev A
Date: Thursday, December 18, 2025	Sheet 11 of 12			

MCULink Voltage Translation Buffers







NXP

ICAP Classification: CP: IUO: X PUBI:

Drawing Title: **FRDM-MW30-S**

Page Title: **MCU_LINK_VCOM_SIO**

Size	Document Number	Rev
C	SCH-95995 PDF: SPF-95995	A

Date: Thursday, December 18, 2025 Sheet 12 of 12